

Caleb Chia

Aspiring physicist, current national serviceman

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About Me

Physics student with experience in academic research and electrical engineering. Multiple olympiad achievements across physics, nuclear science, chemistry, mathematics, and informatics. Experienced in leading technical teams and mentoring students. Currently serving my 2-year national service in the Singapore Armed Forces as an army intelligence specialist.

Education

Anglo-Chinese School (Independent), Singapore 2019–2024
International Baccalaureate Diploma Programme (Nov 2024): 43/45
Graduated: December 2024

Academic Activities

Singapore Physics Olympiad Training Team 2024
Selected for national team, trained at the National University of Singapore. Represented Singapore at the International Nuclear Science Olympiad (INSO 2024) and won a Gold medal.

Engineering Intern, BorgWarner 2024
Short work attachment in electrical engineering department. Worked on circuit design, SPICE-based simulations, RF-testing and bench-testing of devices (mainly high-power inverters).

Research Intern, Nanyang Technological University, School of Electrical and Electronic Engineering 2023–2024
1-year research attachment aimed at optimizing devices for high-power, high-frequency applications. Final report titled: “*Surface Passivation with Al_2O_3/SiN Bilayer in Advanced Gallium Nitride High Electron Mobility Transistors (HEMTs)*,” defended 29 February, 2024.

Executive Committee, ACS(I) Coding Competition Team 2023–2024
Organized and helped develop *Hack@AC 2024*, a national pre-university CTF. Conducted training sessions and represented school in cybersecurity and competitive programming competitions. Explored machine learning on the side.

Vice-President, ACS(I) Mathematics Competition Team 2022–2024
Led internal competitions, mentored juniors and represented school in math competitions.

Non-Academic Activities

Audio-Visual Director, Bukit Panjang Methodist Church 2024–Present
Lead AV technician for youth worship hall. Managed volunteer scheduling, planned system upgrades and mentored junior volunteers.

Class Chairperson, ACS(I)

2024

Planned and organized class activities. Awarded ACS(I) Pastoral Care Award.

Volunteer Chess Teacher, *Project Reperio*

2023–2024

Shared my love for the game with Rohingya children as part of a 1-year holistic education program planned by *Project Reperio*, in collaboration with *Cahaya Surya Bakti*.

Volunteer Tutor, *Heartfriends Student Care Centre*

2023

Managed volunteer scheduling, tutored primary students in math and engaged them with games, as part of a 4-month class project.

Honors & Awards

- Gold Medal, International Nuclear Science Olympiad (2024)
- Gold Medal, Singapore Physics Olympiad (2023)
- Silver Medal, National Olympiad for Informatics (2023)
- Silver Medal, Singapore Chemistry Olympiad (2023)
- Silver Medal, Singapore Mathematical Olympiad (2022–2024)
- Distinction (Most Comprehensive Analysis), Singapore International Mathematical and Computational Challenge (2024)
- Distinction, International Mathematical Modelling Challenge (2024)
- Gold Award, Nanyang Research Programme (2024)
- 3rd Team Placing, Cyberthon (2024)
- ACS(I) Dean's List - Physics, Chemistry, Mathematics (2024)
- MOE Engineering and Tech Programme Scholarship (2023)
- ABRSM Piano Grade 8 Certification (2024)

Standardized Testing**IBDP: 43/45**

November 2024

HL Physics (7), HL Chemistry (7), HL Mathematics AA (7), SL English A: Lang & Lit (6), SL Economics (7), SL Chinese B (7)

SAT: 1570/1600

October 2024

Math: 800, EBRW: 770

Other Interests

Cycling, bouldering, cubing, movies, reading (mainly biographies nowadays)